

Enhancing building energy efficiency with thermal mass optimization

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ABSTRACT

High-density urban buildings contain substantial thermal mass, storing significant energy and offering notable potential for heating energy savings. However, effectively harnessing this energy remains challenging due to the spatiotemporal variability of heat storage–release behavior in building components, which often misaligns with building operational demands. This study reveals that thermal mass tends to store heat when it is not needed and release it when buildings do not require it, especially in cities where some buildings are only occupied during the day and others at night. To address these challenges, this study proposes a novel thermal mass arrangement strategy, derived from extensive real-world data analysis. Significant variations in component thermal behavior across different operational schedules were first identified from data collected in 76 rooms. Subsequently, key factors influencing these variations were pinpointed using stepwise linear regression, informing optimization strategies developed through simulations. These strategies were then validated in cold regions using conduction transfer function models (error margin of 3.6 %), which confirmed their year-round effectiveness for both individual buildings with distinct occupancy patterns and groups of buildings. The results demonstrate that optimizing thermal mass arrangements tailored to specific building schedules can significantly enhance energy efficiency. Contrary to prior research advocating for the sole increase in thermal mass, this study indicates that without strategic guidelines, such measures may exacerbate thermal utilization inefficiencies, complementing existing research on thermal storage materials in buildings. Reducing excess heat storage is shown to be beneficial for daytime-use buildings, while nighttime-use buildings benefit from storing heat for evening use. Adjusting the quantity and orientation of thermal mass, alongside optimizing operational schedules, achieves 4–12 % energy savings, with greater benefits in high-solar-radiation areas.

1. Introduction

The global energy crisis has intensified [1], with the construction industry accounting for 36 % of total energy consumption [2]. Ongoing urbanization increases building density and energy demand, driving a steady rise in annual energy consumption [3]. Heating accounts for nearly 21.2 % of the operational energy use in urban buildings [4], with non-transparent building components contributing to 63.3 % of heat loss [5]. These components (such as external walls, internal walls, and floor slabs) are largely made from high-density concrete [6], serving as thermal mass, absorbing, storing, and releasing heat to regulate indoor temperatures [7]. Urbanization has expanded the scale and density of built environments, increasing the proportion of thermal mass in cities [8]. Recent studies emphasize the considerable heat capacity of thermal mass [9], enabling it to store and release energy [10], providing

flexibility [11], and supporting the building-as-battery (BaB) concept [12] to reduce heating demand [13]. With the growing integration of renewable energy technologies [14], thermal mass further enhances climate resilience and grid flexibility by lowering average and peak loads [15]. These findings highlight the increasing potential to optimize thermal mass utilization in high-density urban areas [16].

1.1. Literature review and research gap

While previous studies have focused on optimizing materials and structures to enhance thermal mass performance for energy savings, researchers have paid limited attention to strategies tailored to varying building operational schedules. Timing and specific energy demands are crucial to the operational effectiveness of thermal mass, yet this area remains underexplored in quantitative analysis and targeted solutions.

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Existing strategies aim to improve heat utilization through integration and manipulation of thermal mass into energy storage systems [17] or advanced materials [18]. This includes optimizing photovoltaic use [19], capturing industrial waste heat [20], reducing peak electricity demand [12], extending preheated energy retention [21], shifting daytime heat to nighttime use [22], balancing thermal comfort and energy savings [23], and converting indoor heat into electricity [24]. However, such approaches are constrained by building use schedules. For instance, daytime-use buildings require immediate heat use rather than storage. Thus, analyzing heat storage-release cycles across building components and schedules is important but remains insufficiently explored in current research.

Developing schedule-specific strategies is challenged by the significant spatial and temporal variability of thermal mass behavior. Existing models predominantly focus on environmental and structural parameters but struggle to explore spatiotemporal variations effectively. Specifically, environmental factors like solar radiation and indoor/outdoor temperatures, considered primary heat sources for thermal mass, fluctuate significantly across locations, complicating consistent optimization. Solar radiation is inherently unstable due to the sun's diurnal cycle, hindering integration with other energy systems [25]. Indoor air temperature impacts energy consumption [26], while outdoor air temperature influences heat release processes [27]. These temperature variations occur within buildings, such as between different rooms [28], affecting heat exchange efficiency [29]. Structural factors, including U-value and heat capacity, influence component thermal behavior [30], varying by location and construction methods [31]. Given these complexities, a spatiotemporal analytical framework integrating environmental and structural dimensions is essential.

Tailoring strategies for buildings with different operational schedules requires large-scale, high-resolution spatiotemporal data from real-world buildings, which remains scarce. Most current studies rely on small-scale experiments or simulations, often based on single rooms, failing to capture the complex dynamics of full-scale buildings. Effective thermal mass systems have to reconcile complex demands, requiring detailed validation under real-world conditions [32]. The implementation also entails maintenance demands and operational uncertainties, necessitating testing in real-building contexts [33]. The lack of testing for new thermal storage materials in buildings has led to risks, including inconsistent energy performance [34] and reduced indoor comfort [35]. Thus, current component-level optimizations or simulations that lack thorough validation against actual energy use patterns [36] limit their applicability in practice.

1.2. Objective and challenges

Existing studies acknowledge the potential benefits of aligning thermal mass heat storage-release cycles with building operational schedules; however, achieving this is nontrivial. First, few studies have quantitatively examined the mismatch between the inherent heat storage-release behaviors of building components and building schedules, and even fewer have proposed viable solutions. The analysis of this issue and the exploration of targeted strategies remain limited. Second, beyond temporal aspects, building components often exhibit diverse thermal behaviors across spatial locations, necessitating a spatiotemporal analytical framework. Yet, most existing research fails to analyze this complexity effectively. Third, given the complex spatiotemporal variations in thermal mass behavior, large-scale, high-resolution real-world data are needed. However, existing research primarily relies on small-scale experiments or simulations, which may not capture the complexity of real-world building dynamics.

In response, this study aims to optimize the heat storage-release behavior of thermal mass to better align with varying building energy demands, thus enhancing energy efficiency. As previously mentioned, existing research has yet to explore these benefits fully. Prior studies typically assume that increasing thermal mass improves indoor thermal

environments [37], but this is not universally applicable. For instance, thermal mass commonly absorbs solar heat during the day and releases it at night (Fig. 1), which conflicts with the needs of daytime-use buildings that require heat during daytime hours in winter. Consequently, increasing thermal mass without strategic guidance can paradoxically lead to higher energy consumption. Therefore, rather than simply increasing thermal mass, it is critical to ensure that stored energy releases align with building demand cycles through targeted design strategies.

1.3. Overview of this study

In response to these challenges, this study first employs real-world data from urban buildings to reveal component thermal variations within 24 h under different operational schedules, highlighting optimization potential. Subsequently, statistical modeling identifies key factors driving these variations, informing the development of strategies aligned with actual energy use patterns. Finally, year-round simulations in cold-region buildings and building groups validate the proposed strategies, achieving energy savings while demonstrating cost-effectiveness and broad applicability. The main contributions of this study are as follows:

1. Leveraging extensive real-world data, this study systematically analyzes the spatiotemporal variations in thermal behaviors of diverse building components and quantifies their impact mechanisms using statistical models.
2. This study proposes, validates, and compares thermal mass arrangement strategies tailored to align component heat storage-release cycles with varying building energy demands, based on both measurements and simulations.
3. This study demonstrates the notable energy-saving benefits and broad applicability of strategies through long-term simulations across multiple representative locations in cold regions.
4. To the best of our knowledge, this study is novel in extending the optimization of thermal mass from isolated components to their spatial distribution, considering diverse building energy demands. Examining the dynamic interactions among building layouts, thermal mass, and energy performance reveals a promising direction for achieving consistent, impactful, and low-cost energy efficiency improvements in individual buildings and building groups.

Overall, this research shifts the focus from individual building components to their collective impact on entire building groups, aiming to fill a void in the literature by examining the spatial and temporal effects of thermal mass. These findings offer valuable insights into optimizing thermal mass layouts for varied building demands, complementing existing research on building materials and structures for greater energy efficiency. The optimization potential highlights the importance of tailored strategies for different energy demands, as a one-size-fits-all approach may increase resource consumption.

The structure of this paper is as follows: **Section 2 introduces the research methods**, including the research framework (2.1), data collection (2.2), and simulation procedures (2.3). **Section 3 presents the results**, focusing on thermal mass spatial and temporal optimization potentials (3.1), impact factor and mechanism analysis (3.2), and validation across cold regions (3.3). **Section 4 discusses the findings**, compares them with existing studies (4.1), analyzes component thermal storage mechanisms (4.2), and presents strategy development (4.3), expansions (4.4), and limitations (4.5). **Section 5 concludes the study.**

2. Method

2.1. Research framework

Fig. 2 illustrates a four-step research framework to align thermal

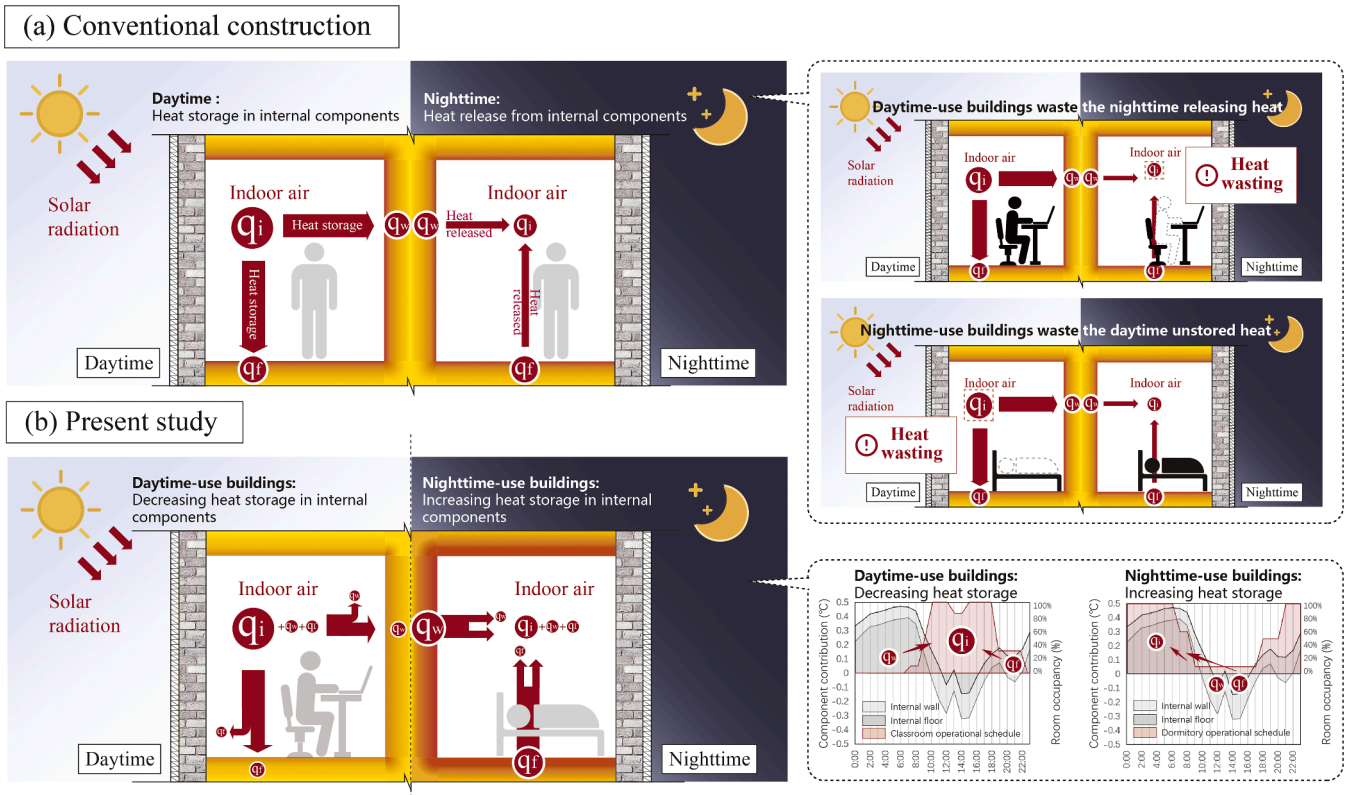


Fig. 1. Thermal mass optimization to harness wasted heat in conventional constructions and improve building energy efficiency: (a) conventional layouts with daytime heat storage and nighttime release, causing heat waste due to mismatched thermal needs; (b) optimized layouts with reduced heat storage for immediate use in daytime-used buildings and enhanced storage for delayed release in nighttime-used buildings, aligned with thermal demand patterns.

* q_i : the heat of the indoor air, q_w : the heat stored in the internal wall, q_f : the heat stored in the floor slab.



Fig. 2. Four-step approach for identifying and validating building layout strategies based on real-world thermal mass behaviors and building energy demands.

mass heat storage-release cycles with dynamic building energy demands. **Step 1: Data collection (Section 2):** This study established a comprehensive measurement dataset, including room dimensions, structural details, and building operational schedules. **Step 2:**

Mechanism analysis (Section 3.1): We used measured data to assess the influence of building components on indoor environments. Newton's law of cooling was applied to quantify heat exchange, while energy management systems monitored building energy demands over time.

Step 3: Strategy development (Section 3.2): Both empirical analysis and simulation were employed to formulate optimization strategies. Stepwise linear regression identified key environmental and structural factors, based on measured data. These findings informed the design of component layout strategies, which were validated in simulations to ensure practical applicability. **Step 4: Multi-scenario testing (Section 3.3):** Year-round simulations were conducted across different building types, locations, and usage scenarios to test the performance of the proposed strategies based on conduction transfer functions. Consequently, grounded in measured data, this framework analyzes component thermal mass mechanisms and develops strategies through numerical methods. Simulations further demonstrate the broad applicability and stability of the proposed strategies. The combined approach confirms that both real-world observations and literature-based analyses were reliable.

Section 2 presents the methodological details. Section 2.2 outlines the measurement process: 2.2.1 introduces climate characteristics of cold regions; 2.2.2 describes the measured buildings; 2.2.3 details the collection of indoor/outdoor environmental data and occupancy information; 2.2.4 analyzes heat transfer between components and indoor air with component contributions; and Section 2.2.5 applies regression models to identify influencing factors for strategy development. Section 2.3 presents the simulation process: 2.3.1 introduces the modeling software and procedures; 2.3.2 ensures the simulation reflects real conditions; 2.3.3 defines validation parameters, integrating strategy inputs derived from real cases; and 2.3.4 quantifies the energy-saving effects of the proposed strategies.

2.2. Measurement: real-world data collection

2.2.1. Information on the measured area

The heating energy consumption across China's vast cold regions [33] presents a significant challenge [38]. Cities in these regions experience considerable variations in solar radiation [39], influenced by differences in direct solar gain [40] and the orientation of building facades. During the heating season, southern facades can receive up to eight times more solar radiation than northern ones, resulting in overheating in southern rooms and insufficient heating in northern rooms, collectively increasing urban heating energy demand. Beijing is selected as the representative city due to its high heating requirements and substantial urban development [40]. The city covers approximately 3517 km², with about 780 km² of heated area consuming around 100 TWh of heat during the heating season [41]. Beijing's climate conditions are representative, as detailed in Appendix B.5, making it a suitable case for analyzing thermal mass strategies in urban heating.

2.2.2. Information on the measured buildings

To ensure the reliability of the real-world data analysis, measurements were taken in 76 rooms across 18 buildings, which accounted for variations in orientation, U-value, areal heat capacity (km), room dimensions, and operational schedules. The details of these conditions are provided in Appendix B.1 and B.2. This comprehensive dataset allows for the identification of patterns, correlations, or disparities in the statistical analysis [42]. Table 1 and Fig. 3 provide detailed information on these rooms.

Table 1
Measured rooms and their characteristics.

Building	Rooms dimension [mm]	Component	U-value [W/m ² ·K]	km [kJ/m ² ·K]	Building	Rooms dimension [mm]	Component	U-value [W/m ² ·K]	km [kJ/m ² ·K]
D-1	A = 17.71 m ² W = 3850 H = 3050	External wall	0.33	256	T-1	L = 53.25 m ² W = 7250 H = 3300	External wall	0.82	1620
		Internal wall	3.46	276			Internal wall	2.12	485
		Floor	3.60	230			Floor	1.74	506
D-2	L = 13.5 m ² W = 3000 H = 3200	External wall	0.33	256	T-2	A = 5.52 m ² W = 1900 H = 3500	External wall	1.50	790
		Internal wall	3.46	276			Internal wall	1.50	790
		Floor	3.60	230			Floor	1.75	460
D-3	A = 11.69 m ² W = 2750 H = 3350	External wall	0.42	475	T-3	A = 40.35 m ² W = 5300 H = 2800	External wall	1.90	550
		Internal wall	2.14	378			Internal wall	1.90	550
		Floor	3.27	345			Floor	1.74	644
D-4	A = 13.5 m ² W = 3000 H = 2750	External wall	1.67	690	T-4	A = 83.62 m ² W = 7400 H = 3150	External wall	1.37	882
		Internal wall	1.94	454			Internal wall	2.06	504
		Floor	2.76	217			Floor	1.74	575
D-5	A = 21.00 m ² W = 3500 H = 2800	External wall	1.67	690	T-5	A = 168 m ² W = 14,800 H = 3150	External wall	1.37	882
		Internal wall	1.94	454			Internal wall	2.06	504
		Floor	2.76	217			Floor	1.74	575
D-6	A = 16.83 m ² W = 3300 H = 3100	External wall	1.64	693	T-6	A = 91.76 m ² W = 11,470 H = 3500	External wall	0.28	299
		Internal wall	2.06	504			Internal wall	0.60	372
		Floor	1.74	690			Floor	3.27	345
D-7	A = 13.5 m ² W = 3000 H = 2600	External wall	0.36	252	T-7	A = 61.60 m ² W = 6280 H = 3500	External wall	0.85	1550
		Internal wall	2.42	400			Internal wall	0.85	1550
		Floor	1.74	460			Floor	1.75	550
D-8	A = 26.63 m ² W = 3750 H = 2850	External wall	0.52	203	T-8	A = 39.38 m ² W = 4500 H = 3500	External wall	1.90	560
		Internal wall	2.28	430			Internal wall	2.75	315
		Floor	1.74	414			Floor	1.75	368
T-10	A = 20.95, 46.55 m ² W = 3150, 7000 H = 3850	External wall	1.19	529	T-9	A = 21.56, 44.28 m ² W = 3750, 7700 H = 3150	External wall	0.45	163
		Internal wall	1.58	353			Internal wall	2.40	306
		Floor	3.68	207			Floor	3.46	276

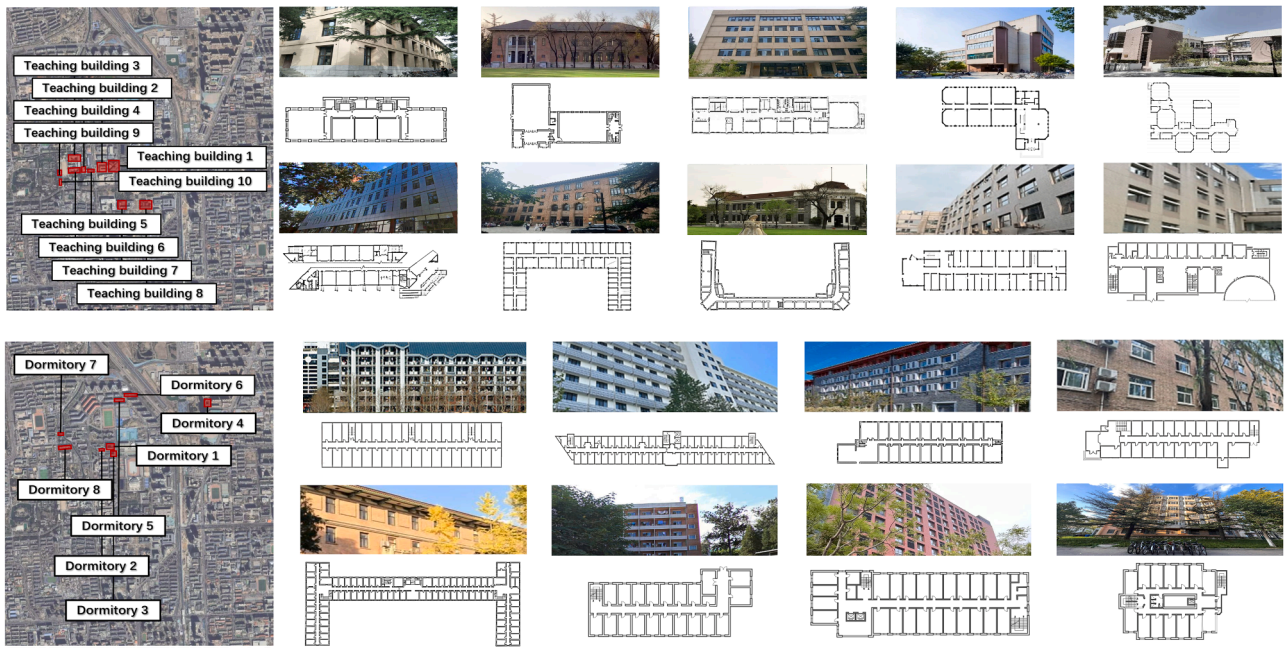


Fig. 3. Façades and layouts of the investigated buildings.

2.2.3. Measurement method

As shown in Fig. 4, data collection included meteorological parameters, indoor air temperature, and surface temperatures of building components. The measurement approach builds on sensor deployment strategies from prior studies [43], extending them to diverse architectural configurations to support detailed spatiotemporal heat storage and release analysis. All measurements specified in Fig. 4 were recorded synchronously at five-minute intervals to capture dynamic thermal variations. During the non-heating season, under low outdoor temperatures, this study conducted measurements for over one month to reflect cold-season conditions. This study collected data in unoccupied rooms to avoid interference from internal heat sources, preserving the natural thermal response of building components [44]. Meteorological data were recorded using a YGY-CJY4 weather station installed on a rooftop representative of local conditions [45]. Indoor air temperature was measured using a CEM DT-171 sensor, which was placed 0.6 m above the floor at the center of the room, avoiding direct sunlight, to reflect the occupant-level environment [28]. The surface temperatures of building components were measured using a WZY-1 sensor at five representative points per surface. Sensor placements in Fig. 4 show the orange dots indicating standard components and the blue dots representing irregular components. Data processing followed Eqs. (1–3) [46].

$$t_{\text{regular component}} = (t_1 + t_2 + t_3 + t_4 + t_5)/5, \quad (1)$$

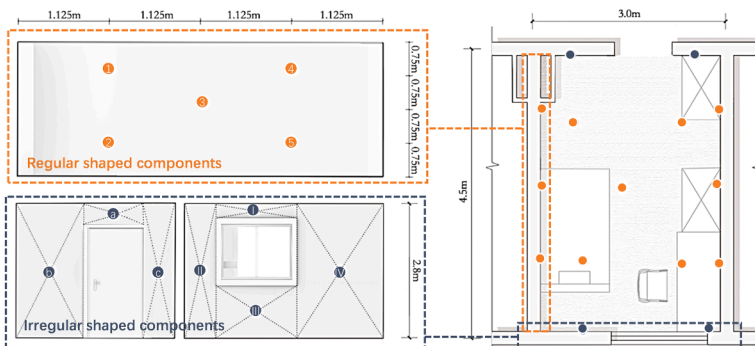


Fig. 4. Sensor specifications and measurement arrangements of building component surface temperatures.

$$t_{\text{internal wall}} = (S_a * t_a + S_b * t_b + S_c * t_c) / (S_a + S_b + S_c), \quad (2)$$

$$t_{\text{external wall}} = (S_I * t_I + S_{II} * t_{II} + S_{III} * t_{III} + S_{IV} * t_{IV}) / (S_I + S_{II} + S_{III} + S_{IV}), \quad (3)$$

$t_{\text{regular component}}$, $t_{\text{internal wall}}$, $t_{\text{external wall}}$: Component inner surface temperature, °C,

t_1, t_2, t_3, t_4, t_5 : 5 points temperature of regular components inner surface, °C,

$t_a, t_b, t_c, t_I, t_{II}, t_{III}, t_{IV}$: Temperature of irregular components inner surface, °C,

$S_a, S_b, S_c, S_I, S_{II}, S_{III}, S_{IV}$: Represented area of the corresponding temperature measurement point, m².

Building operational schedules were obtained from online systems, as shown in Fig. 5. The schedules of teaching buildings represent daytime-use operations, while energy usage data from dormitories reflect nighttime-use operations. Hourly occupancy rates were obtained from course schedules extracted from the teaching management system over one year. Dormitory occupancy patterns were estimated using hourly electricity consumption data from the same period, following the method detailed in Appendix B.6 from previous studies [47].

Outdoor environment			
Measured parameter	Brand and Model	Measuring range	Accuracy
Surface horizontal radiation	YGJ-CJY4 meteorological station	0 to 2000 W/m ²	1 W/m ² (±0.3%)
Outdoor air temperature		-40 to 100 °C	0.1 °C (±0.5 °C)

Indoor environment			
Measured parameter	Brand and Model	Measuring range	Accuracy
Indoor air temperature	CEM DT-171 datalogger	-40 to 70 °C	0.1 °C (±1 °C)
Components temperature	WZY-1	-50 to 100 °C	0.1 °C (±0.3 °C)

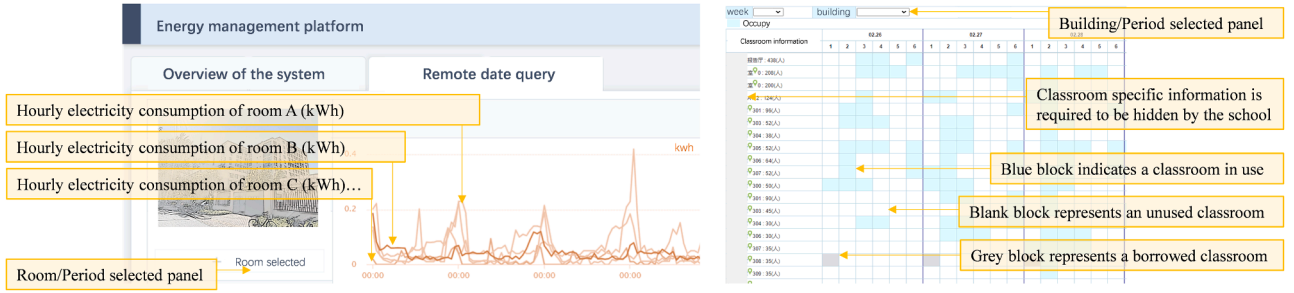


Fig. 5. Building occupancy rate analysis: online management platform for dormitories and classrooms.

2.2.4. Measured data analysis: component contribution

This study used Eq. (4) [48] to calculate heat flow and assess the component contribution. The result of Eq. (4) > 0 indicates heat transfer from the component to the indoor air, implying a positive impact during winter. Conversely, the result of Eq. (4) < 0 signifies a negative impact. Eq. (4) is presented below:

$$E_{total} = \alpha \cdot \int_{\tau} [t_{component}(\tau) - t_{indoor-air}(\tau)] d\tau, \quad (4)$$

E_{total} : heat transmission across the components to the indoor air, kWh/m²,

α : the internal surface heat transfer coefficient. According to Chinese standard GB 50176–2016 [49], the thermal properties of the material remain largely stable over time, with a value of 8.7 W/(m²·°C),
 $t_{component}(\tau)$: the τ moment components inner surface temperature, °C,
 $t_{indoor-air}(\tau)$: the τ moment indoor air temperature, °C.

2.2.5. Measured data analysis: modeling the impact mechanism of component contribution

To assess the influence of various factors on building component contributions, stepwise linear regression models were applied, as detailed in Appendix B.4. This data-driven approach is widely utilized in research due to its interpretability [29], particularly for modeling relationships between dependent and multiple independent variables [42]. The modeling procedure involved (a) conducting univariate regression to identify significant predictors (P-value < 0.05) related to component contributions. (b) incorporating these predictors into the model through stepwise regression based on their R² values. (c) constructing the final model with selected predictors while ensuring that the variance inflation factor (VIF) values remained below 10, indicating no significant multicollinearity [50]. The final model, shown in Eq. (5), is derived from stepwise linear regression after selecting k independent variables from the potential predictors:

$$\hat{\beta} = \underset{\beta \in \mathbb{R}^k}{\operatorname{argmin}} \|Y - X\beta\|^2 = \underset{\beta \in \mathbb{R}^k}{\operatorname{argmin}} \sum_{i=0}^m \left(y_i - \sum_{n=0}^{k-1} \beta_n x_{in} \right)^2, \quad (5)$$

$\beta_0, \beta_1, \beta_2, \dots, \beta_{k-1}$: the impact factors for the environmental and structural factors,

$x_0, x_1, x_2, \dots, x_m$: the values of the environmental and structural factors,

$y_0, y_1, y_2, \dots, y_m$: the values of the component contribution.

2.3. Simulation: validation of strategies across time periods and locations

2.3.1. Information on the simulation process

The proposed strategies were validated using DesignBuilder (version 7.0.0.096), which operates on EnergyPlus (version 9.4.0) developed by the U.S. Department of Energy. Heat absorption, storage, and release in

building components were modeled using conduction transfer functions (CTFs), as detailed in Appendix B.3. DesignBuilder has been validated against ASHRAE Standard 140P and BEPAC benchmarks for thermal simulation accuracy [30]. This modeling approach quantitatively evaluates strategy effectiveness under various temporal and spatial conditions. The core CTF method, based on the heat balance principle, is expressed in Eq. (6) [51]:

$$q_{i,t} = \sum_{m=1}^M X_{k,m} t_{i,t-m+1} - \sum_{m=1}^m Y_{k,m} t_{o,t-m+1} + \sum_{m=1}^k q_{i,t-m} F_m, \quad (6)$$

$q_{i,t}$: the heat at the interior surface of the component at simulation time step t , W/m²,

$t_{i,t-m+1}$: the inside surface temperature at the time step t and a fixed number of previous steps, °C,

$t_{o,t-m+1}$: the outside surface temperature at time step t and a fixed number of previous time steps, °C,

$q_{i,t-m}$: the heat flux at the component interior surface at a fixed number of previous time steps, W/m²,

$X_{k,m}, Y_{k,m}, F_m$: the conduction transfer functions, which are constant for a particular building element over the entire simulation, W/(m²·°C).

Load calculations rely on zone heat balances, requiring energy balances for accurate thermal zone simulation. Heat conduction links two of these energy balances, leading to the following load calculation methodology in Eq. (7):

$$C_z \frac{dt_z}{dt} = q_{conv} + \sum_{i=1}^{surfaces} h_i A_i (t_i - t_z) + \dot{m}_{infCp} (t_{\infty} - t_z) + \sum_{k=1}^{zones} \dot{m}_{Cp} (t_k - t_z) + q_{system}, \quad (7)$$

t_{∞} : the outdoor dry bulb temperature, °C,

t_z : the zone mean air temperature, °C,

q_{conv} : the total convective heat transfer from internal sources, W/m²,
 $\sum_{i=1}^{surfaces} h_i A_i (t_i - t_z)$: convective heat transfer from all zone surfaces, W/m²,

$\dot{m}_{infCp} (t_{\infty} - t_z)$: sensible heat transfer due to infiltration of outside air, W/m²,

$\sum_{k=1}^{zones} \dot{m}_{Cp} (t_k - t_z)$: the sensible heat transfer to mix air with other zones, W/m²,

q_{system} : the output of a conventional system, W/m²,

C_z : the product of the mass of the zone air and the specific heat of air, J/K.

2.3.2. Information on the calibration process

As detailed in Appendix B.7, model calibration involved (a) collecting meteorological and indoor air temperature data using YGY-CJY4

and CEM DT-171 sensors (Fig. 4) in an unoccupied, non-HVAC room in Building D-2 (Table 1); (b) calibrating the simulation model with this dataset, as shown in Fig. 6; and (c) conducting strategy validation simulations using the China Standard Weather Data (CSWD) and standardized HVAC settings, with simulated energy consumption closely matching benchmark values from GB 55015–2021. The model in Fig. 6 achieved a mean bias error (MBE) of 1.9 % and a coefficient of variation of root mean square error (CV (RMSE)) of 3.6 %, both within the ASHRAE Guideline 14–2014 thresholds [52], demonstrating its accuracy in capturing real thermal behavior. Overall, the real-world measurements verified the accuracy of the thermal mass behavior modeling, while the simulation outcomes confirmed the reliability of energy consumption predictions under standardized conditions.

2.3.3. Information on the validation parameters

The validation data for meteorological conditions and architectural strategies align with standard requirements. The architectural strategy involves modifying internal room dimensions while preserving the external structure and boundaries of the buildings. Parameters were derived from real-world building datasets on ArchDaily, ensuring they reflect the latest architectural designs. Modifications included adjusting the quantity and orientation of internal walls and floor slabs, as shown in Fig. 7.

2.3.4. Simulation analysis: energy-saving effects

This study quantifies the energy-saving effects with Energy usage intensity (EUI) [53], calculated by Eq. (8) [29]. The energy-saving rate (ESR) is determined using Eq. (9) [54], while Eq. (10) evaluates the overall energy-saving performance of the building group. Eqs. (8–10) show the calculation methods:

$$EUI_{\text{saving}} = EUI_{\text{initial}} - EUI_{\text{optimization}}, \quad (8)$$

$$ESR = \left[1 - \left(\frac{EUI_{\text{optimization}}}{EUI_{\text{initial}}} \right) \right] \times 100\%, \quad (9)$$

$$ESR_{\text{building group}} = \left[\frac{(S_1 * ESR_1 + S_2 * ESR_2 + \dots + S_n * ESR_n)}{(S_1 + S_2 + \dots + S_n)} \right] \times 100\%, \quad (10)$$

EUI_{saving} : the strategy energy saving intensity, kWh/m²,
 EUI_{initial} : the initial plan energy usage intensity, kWh/m²,
 $EUI_{\text{optimization}}$: the optimization plan energy usage intensity, kWh/m²,
 ESR : the strategy energy saving rate, %,
 $ESR_{\text{building group}}$: the strategy energy saving rate of the building groups, %,
 $S_1, S_2, \dots, S_n$: Represented area of the buildings with different schedules, according to Standard 191–2018 [55] in this study, m².

3. Results

The results are presented in three parts: Section 3.1 identifies thermal mass optimization potential by analyzing measured component contributions and building energy demand patterns. Section 3.2 explores the mechanisms driving these measured contributions and develops corresponding optimization strategies, which are then validated through simulation. Section 3.3 applies the proposed strategy to broader scenarios using long-term simulations to verify their effectiveness.

3.1. Identification of optimization potential by comparing building energy demand and component contributions using measured data

Fig. 8 illustrates the 24-hour operational schedules of teaching buildings and dormitories and the respective thermal contributions of external walls, internal walls, and floor slabs. As shown in Fig. 8(a), teaching buildings operate primarily during the day, while dormitories represent nighttime-use patterns. Fig. 8(b) reveals that component performance varies significantly by time and location. From 10:00 to 22:00 (daytime), most components have a negative impact on the indoor thermal environment, whereas, between 22:00 and 10:00 (nighttime), internal walls and floor slabs exhibit positive thermal contributions. External walls maintain a negative impact throughout the day. Comparing Fig. 8(a) and (b), the alignment between dormitory schedules and periods of positive component contributions suggests greater compatibility, while teaching buildings often coincide with periods of negative impact. These results indicate a mismatch between component behavior and building schedules in some instances, underscoring the need for tailored optimization strategies.

Fig. 9 analyzes component performance to identify cases with high optimization potential. Fig. 9(a) shows that external walls consistently contribute negatively, with contributions dropping to between 0 °C and −3.5 °C in approximately 90 % of observations. Fig. 9(b) indicates that internal walls perform positively at night, with gains (up to 1 °C) in 90 % of cases, but are negative (up to −3.5 °C) in 50 % of daytime cases. Fig. 9(c) demonstrates that floor slabs contribute positively in 90 % of cases at night and 60 % during the day, with the most efficient cases exceeding 1 °C in both periods. Despite these favorable trends, some rooms show minimal or adverse thermal mass performance in Fig. 9(b) and (c). Internal wall contributions range from −1.5 °C to −3.5 °C in low-performing cases, while floor slabs range from −1.1 °C to −2.2 °C. In summary, external walls generally exert a consistent negative influence on indoor thermal conditions. In contrast, internal walls and floor slabs exhibit both positive and negative effects on the indoor environment, demonstrating the potential for enhancing building energy efficiency.

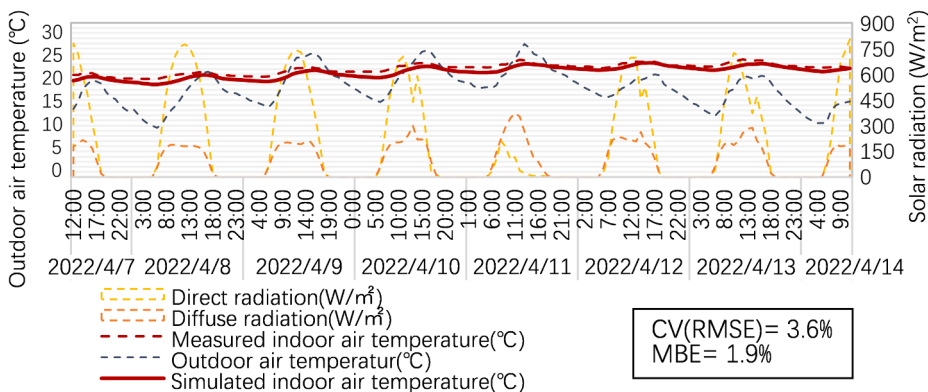
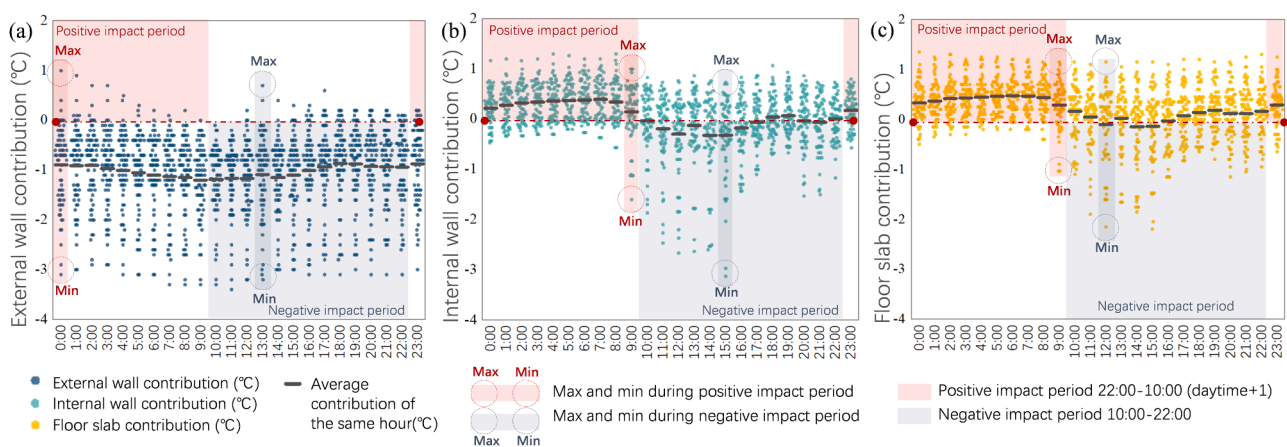
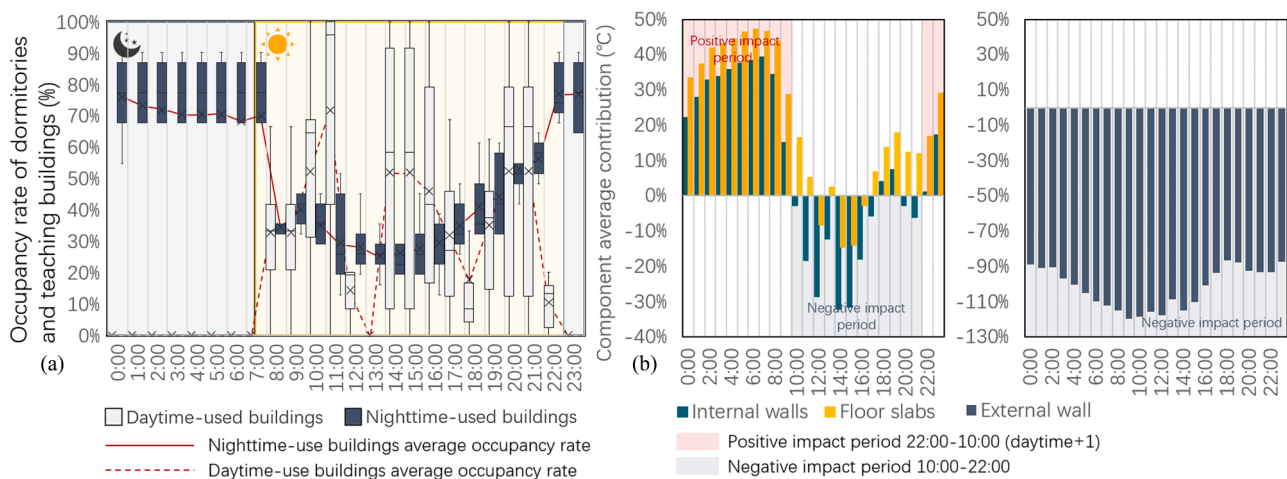
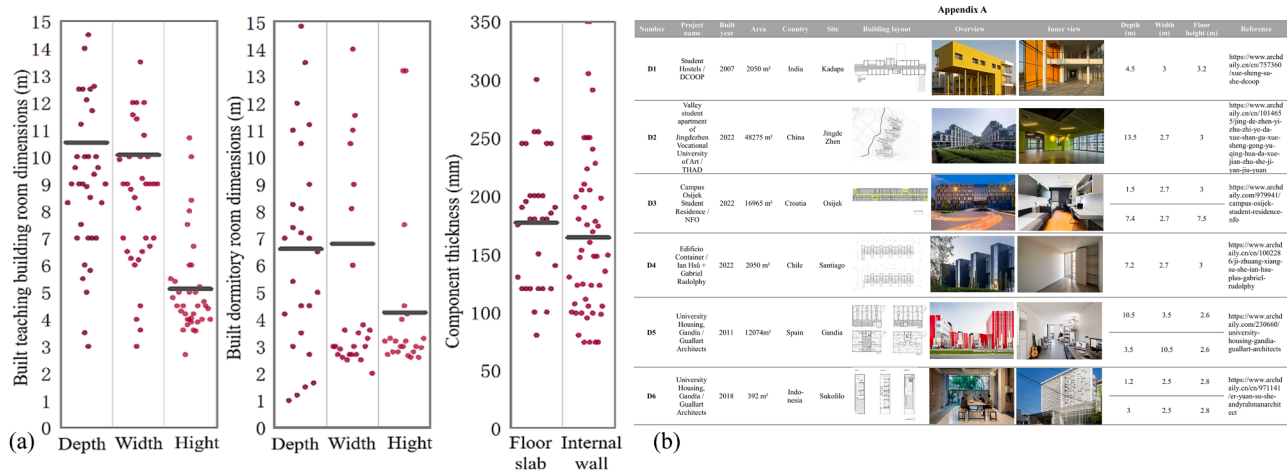


Fig. 6. Calibration results of simulation reliability.



3.2. Mechanism analysis and strategy development from building component contributions based on measurements and simulations

Previous analyses identified general trends and notable cases in

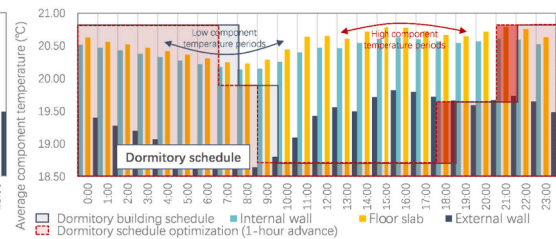
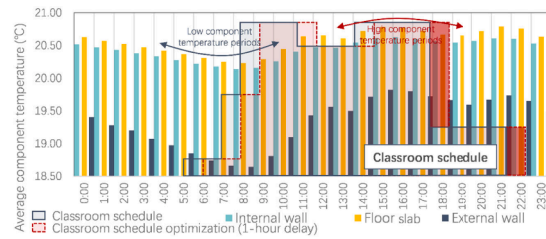
building component contributions during different periods; however, effective optimization requires a deeper understanding of underlying mechanisms. As summarized in [Table 2](#), models with R^2 values ranging from 0.28 to 0.77 effectively capture these processes, indicating that

Table 2

Real-world data on component contributions and impact factors across different time periods.

		Negative impact period 10:00 to 22:00 (Daytime)						Positive impact period 22:00 to 10:00 (Nighttime)					
		Northern room			Southern room			Northern room			Southern room		
		External wall	Internal wall	Floor slab	External wall	Internal wall	Floor slab	External wall	Internal wall	Floor slab	External wall	Internal wall	Floor slab
Data information	Min	-2.70	-3.13	-2.20	-3.30	-2.67	-2.15	-2.70	-3.13	-2.20	-3.30	-2.67	-2.15
	Max	0.20	0.83	1.30	0.70	1.07	1.05	0.40	1.17	1.35	1.00	1.30	1.20
	Std. deviation	0.59	0.52	0.45	0.76	0.62	0.64	0.58	0.49	0.45	0.85	0.58	0.58
	Range	2.90	3.97	3.50	4.00	3.73	3.20	3.10	4.30	3.55	4.30	3.97	3.35
	Number of samples	1388	1388	1388	717	717	717	839	839	839	1036	1036	1036
Environmental factors	Outdoor air temperature (°C)	0.12**	—	—	0.22**	—	—	0.15**	—	—	0.29**	—	—
	Indoor air temperature (°C)	-0.34**	-0.81**	-0.55**	-0.44**	-0.33**	-0.65**	-0.49**	-0.94**	-0.57**	-0.44**	-0.38**	-0.54**
	Solar radiation (W/m ²)	-0.28**	-0.12**	—	-0.17**	-0.25**	-0.15**	-0.25**	-0.14**	-0.10**	-0.15**	-0.24**	-0.17**
Structural and spatial factors	External wall	U-value (W/m ² ·K)	-0.59**($\downarrow$)	-0.07**($\downarrow$)	-0.57**($\downarrow$)	-0.87**($\downarrow$)	—	—	-0.95**($\downarrow$)	-0.43**($\downarrow$)	-0.12**($\downarrow$)	—	—
		κm (kJ/m ² ·K)	2.13**($\uparrow$)	0.28**($\uparrow$)	1.10**($\uparrow$)	0.16**($\uparrow$)	—	—	1.16**($\uparrow$)	0.39**($\uparrow$)	0.42**($\uparrow$)	—	—
	Internal wall	Width/volume (m/m ³)	—	0.68**($\uparrow$)	—	—	—	—	—	0.62**($\uparrow$)	—	—	—
		κm (kJ/m ² ·K)	-1.85**($\downarrow$)	—	-1.15**($\downarrow$)	—	0.23**($\uparrow$)	—	—	—	—	0.10**($\uparrow$)	—
	Floor slab	Height/volume (m/m ³)	—	—	-0.55**($\downarrow$)	—	—	0.13**($\uparrow$)	0.31**($\uparrow$)	—	-0.10**($\downarrow$)	—	-0.15**($\downarrow$)
		κm (kJ/m ² ·K)	1.62**($\uparrow$)	—	—	-0.57**($\downarrow$)	-0.26**($\downarrow$)	—	-0.38**($\downarrow$)	—	—	—	—
R ²		0.77	0.63	0.44	0.50	0.42	0.49	0.71	0.62	0.35	0.42	0.28	0.46
Adjusted R ²		0.77	0.63	0.44	0.50	0.42	0.48	0.71	0.62	0.34	0.42	0.28	0.45

Adjust operational schedules for classrooms and dormitories



Symbols: ** indicates $P < 0.001$; * indicates $P < 0.01$; Terms: "Width, Height/volume (m/m³)" represent the ratio of the room's width and height to its volume, respectively. ($\uparrow$) and ($\downarrow$) represent positive and negative influences, respectively, on the contributions of internal walls and floor slabs, supporting thermal mass optimization strategies; As the findings for external walls (uncolored ($\uparrow$) and ($\downarrow$)) align with existing studies, further discussion is omitted in this paper.

Symbols: ** indicates $P < 0.001$; * indicates $P < 0.01$; Terms: "Width, Height/volume (m/m³)" represent the ratio of the room's width and height to its volume, respectively.

($\uparrow$) and ($\downarrow$) represent positive and negative influences, respectively, on the contributions of internal walls and floor slabs, supporting thermal mass optimization strategies;

As the findings for external walls (uncolored ($\uparrow$) and ($\downarrow$)) align with existing studies, further discussion is omitted in this paper.

environmental factors primarily drive heat exchange, while structural and spatial factors inform optimization strategies for varying building energy demands.

Indoor air temperature is the primary driver of component heat exchange, while solar radiation and outdoor temperature significantly influence heat gains and losses, as shown in Table 2. During the daytime, components such as southern external walls (Indoor air temperature factor: $-0.44 < -0.34$), northern internal walls ($-0.81 < -0.33$), and southern floor slabs ($-0.65 < -0.55$) absorb the most indoor air heat. At night, northern external walls ($-0.49 < -0.44$), internal walls ($-0.94 < -0.38$), and floor slabs ($-0.57 < -0.54$) dominate indoor heat loss. Higher outdoor temperatures improve component performance (Outdoor air temperature factor: 0.12 to 0.29), especially for southern external walls ($0.12 < 0.22$, $0.15 < 0.29$). Solar radiation raises indoor air temperature, leading to greater component heat absorption but reducing their overall contribution, with northern external walls (Solar radiation factor: $-0.28 < -0.17$, $-0.25 < -0.15$) dissipating the most heat, followed by southern internal walls ($-0.25 < -0.12$, $-0.24 < -0.14$) and southern floor slabs ($-0.17 < -0.10$, -0.15), which store significant heat. Overall, southern external walls, floor slabs, and northern internal walls absorb the most indoor air heat during the day, while northern components dominate indoor air heat loss at night. Higher outdoor temperatures benefit the performance of external walls, and solar radiation mainly affects southern internal walls and floor slabs.

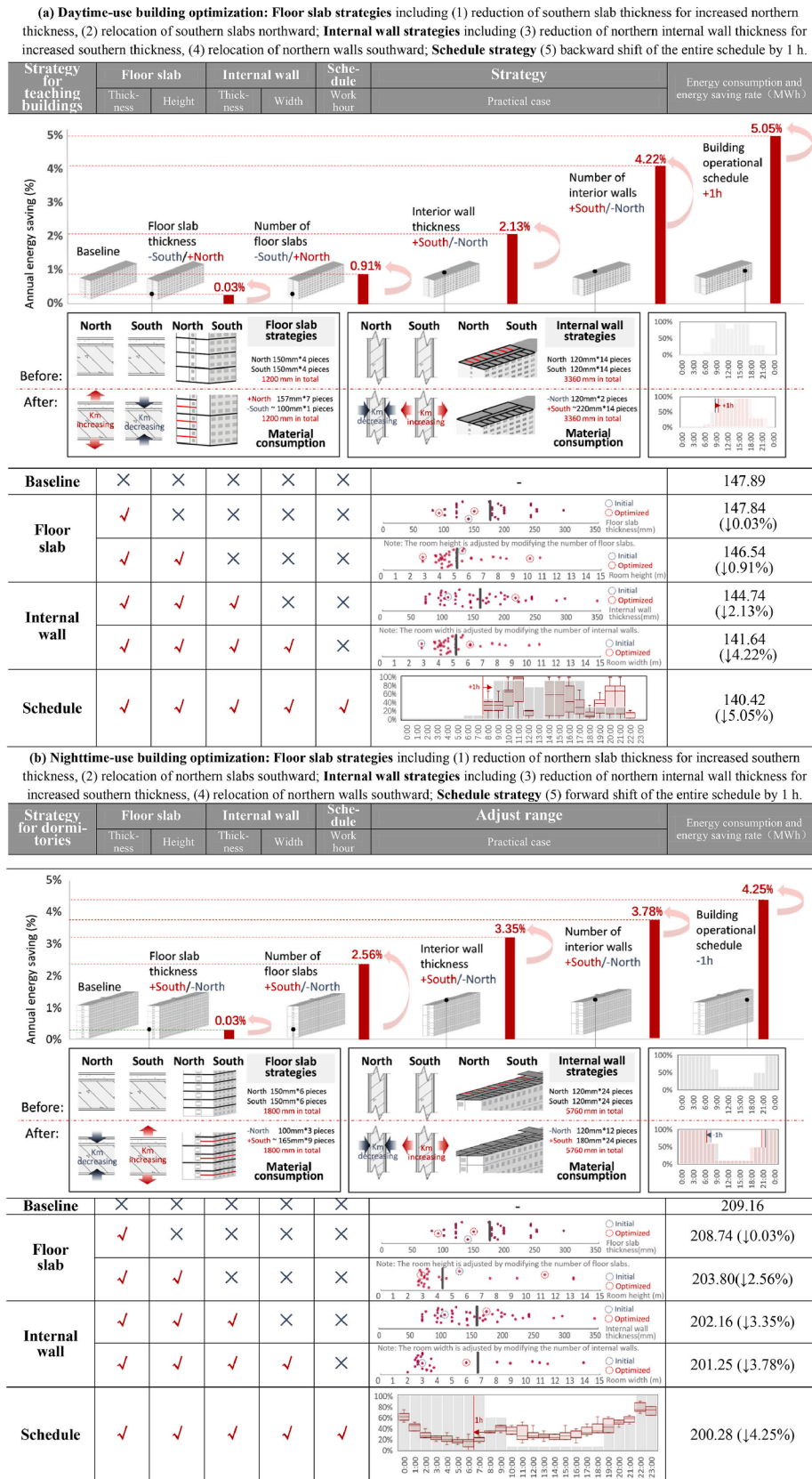
Structural and spatial factors further guide optimization in Table 2. For the external wall, improving the U-value (U-value factor: -0.58 to

$-0.95 \downarrow$) and thermal mass (κm factor: 0.16 to 2.13 $\uparrow$) enhances performance across schedules. Internal walls perform more efficiently in wider northern rooms (width/volume factor: 0.68 $\uparrow$, 0.62 $\uparrow$) with fewer northern internal walls (κm : $-1.85 \downarrow$, $-1.15 \downarrow$), which can be replaced by thicker southern internal walls (κm : 0.23 $\uparrow$, 0.10 $\uparrow$). For floor slabs, daytime-use buildings benefit from lower northern room heights (height/volume: $-0.55 \downarrow$) and thicker northern slabs (κm : 1.62 $\uparrow$), while the reverse applies for nighttime buildings. Aligning building schedules with component performance offers additional benefits. Delaying teaching building schedules from 6:00–22:00 to 10:00–22:00 better aligns with higher component temperatures, potentially reducing heating demand. Advancing dormitory schedules from 22:00–8:00 to earlier hours avoids colder periods between 22:00 and 10:00. In conclusion, these strategies provide guidance for optimizing building layouts, selecting component constructions, and adjusting schedules to improve the building component contributions for different building energy demands.

Table 3 illustrates that the proposed strategies reduce heating energy consumption by 5.05 % in teaching buildings and 4.25 % in dormitories without altering construction materials. In dormitories, shifting thermal mass southward increases daytime heat storage for nighttime use. Strategies include thickening southern floor slabs, reducing southern room height, and thinning northern floor slabs while increasing northern room height, yielding 0.03–2.56 % energy savings. Optimizing internal wall configurations adds another 3.35–3.78 % savings. Schedule adjustments contribute up to a 4.25 % total improvement by better aligning with heat storage cycles. Teaching buildings require different

Table 3

Thermal mass arrangement strategies drawn from measured data analysis for (a) daytime- and (b) nighttime-use buildings in 5-step simulations.



strategies. Southern floor slabs tend to trap excessive heat; thus, strategies such as thickening northern slabs, thinning southern ones, and relocating slabs northward can achieve 0.03–0.91 % in energy savings. For internal wall optimization, thickening the southern wall and reducing the northern wall volume adds 2.13–4.22 % in savings. Scheduling adjustments further reduce heating demand by 5.05 % overall. In summary, different building operational schedules require tailored thermal mass strategies. For daytime-use buildings, heat storage during occupied periods should be minimized, whereas for nighttime-use buildings, daytime heat storage should be enhanced to support nighttime heating demands.

3.3. Evaluation of the energy-saving performance for the strategy across buildings in cold regions over a full year through simulations

Buildings with different operational schedules require tailored thermal mass strategies, while location and building scale variations further influence their effectiveness. As shown in Fig. 10(a), cities with higher solar radiation, such as Lhasa (148 W/m^2 , 1.6°C) and Xigaze (168 W/m^2 , -2.8°C), achieve the highest energy savings of 7–8 %. In contrast, at different building scales, applying inappropriate strategies can lead to considerable energy losses. For example, Fig. 10(b) illustrates a 2 % energy loss in Beijing when daytime-use building strategies are misapplied to nighttime-use dormitories, while teaching buildings that apply inappropriate strategies show a 5 % discrepancy in energy savings. At the campus scale, inappropriately tailored strategies can reduce energy efficiency by 4–12 %. These findings underscore the importance of applying building- and scale-specific strategies to enhance energy efficiency in both individual and grouped buildings, with the most significant benefits observed in cities with higher solar radiation.

Although designed primarily for heat preservation in winter, these strategies demonstrate year-round effectiveness. In Fig. 11(a), building

groups achieve 5–10 % energy savings across various cities during the heating season, with minimal negative impact in summer. Dormitory strategies in Fig. 11(b) yield about 5 % energy savings by raising all-day temperatures. Fig. 11(c) shows that teaching buildings achieve 10–20 % savings during the heating season, with higher indoor temperatures during transitional periods. In conclusion, applying tailored thermal mass strategies yields significant year-round energy savings.

4. Discussion

4.1. Contributions of the present study compared to previous studies

While previous studies have focused on material and structural innovations through small-scale experiments, this study uses real-scale measurements to analyze thermal mass distribution placement, providing guidance for layout design during the building design stage. For example, existing studies, such as [56,57] and [58] in Appendix C.18, explored enhancing southern internal walls with advanced materials to improve heat storage. This study further investigates the benefits of internal thermal mass in different spatial locations, including floor slabs, which are challenging to explore in smaller-scale experiments. Further analysis of component contributions and building energy demands reveals that thermal mass effectiveness varies with occupancy patterns. Tailoring strategies to specific energy demands enhances the energy-saving benefits. Specifically, the present study finds that the energy-saving potential of the improved thermal mass is more pronounced in buildings primarily used at night than those used during the day. Therefore, some existing strategies may be suboptimal in multi-functional buildings, and integrating the proposed strategies could enhance efficiency. For instance, [59,60], and [61] in Appendix C.18, which increase wall thermal mass to capture solar heat, may be more suitable for nighttime-use buildings.

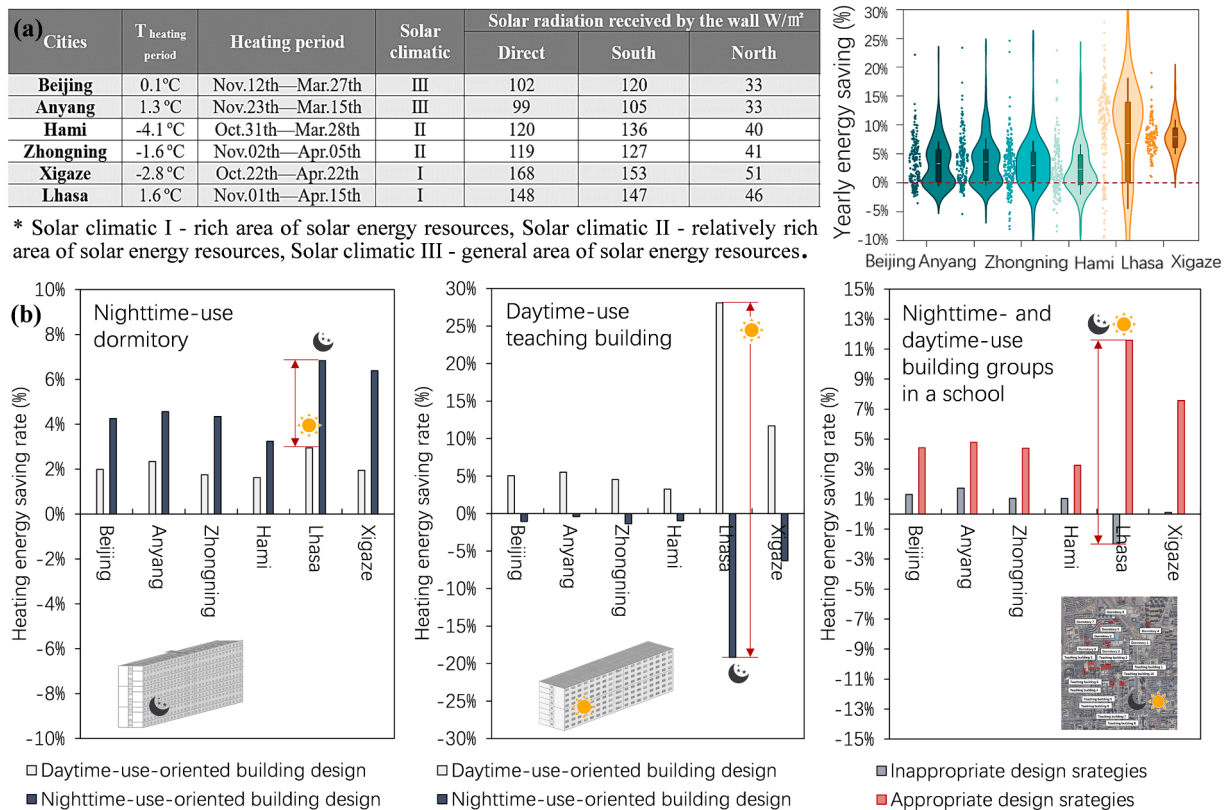


Fig. 10. Simulation results of the proposed strategy in representative cities: (a) climate conditions and annual energy savings; (b) comparison of appropriate and inappropriate arrangements for individual and grouped buildings.

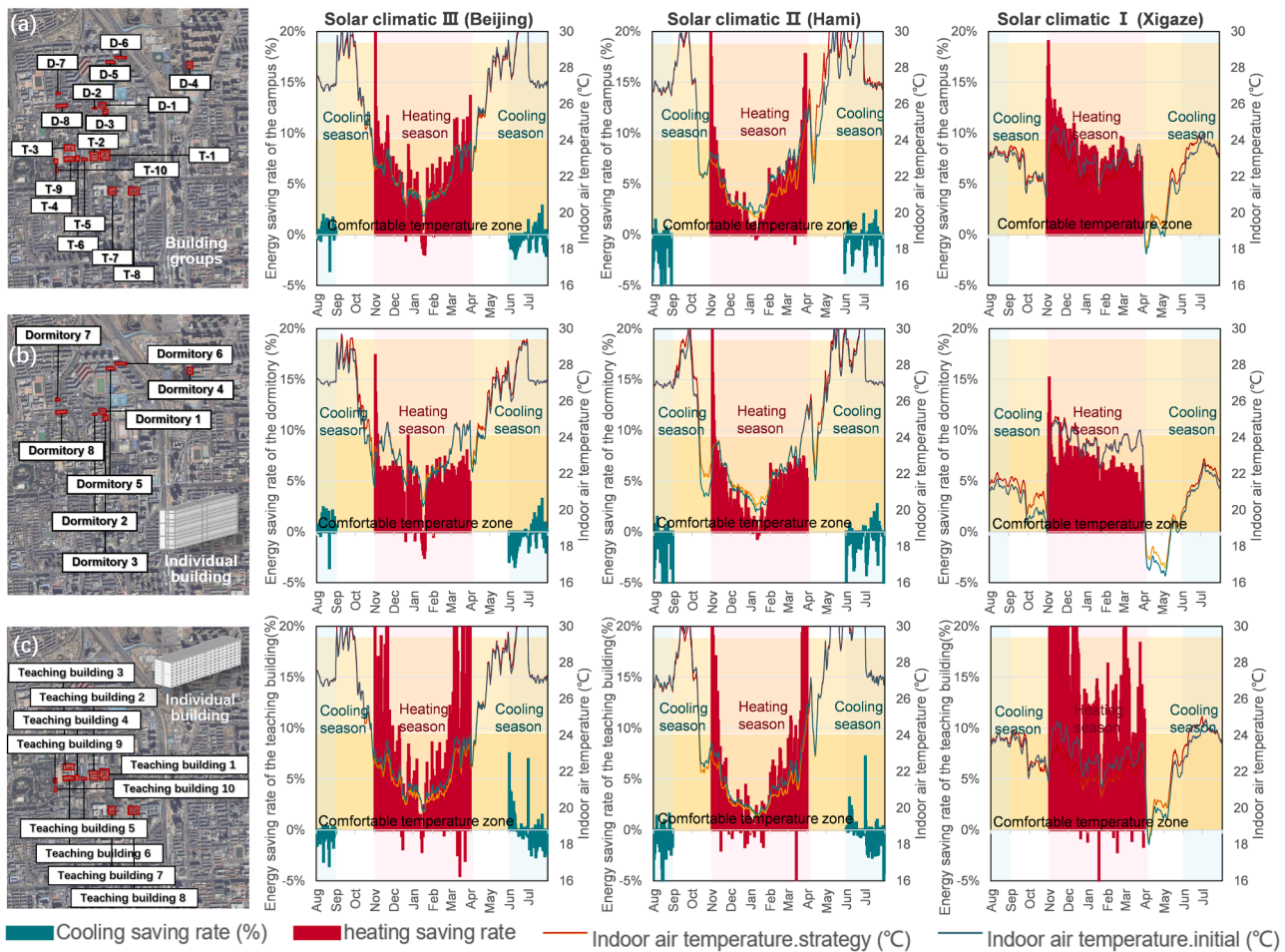


Fig. 11. Simulation results of daily energy-saving effects of the proposed strategy in representative cities, with notable savings during the heating period: (a) building groups, (b) dormitories, and (c) teaching buildings.

4.2. Spatiotemporal mechanisms of thermal storage in real buildings

This study advances the understanding of thermal mass performance across spatial locations and operational periods in real-scale buildings. While previous research relied on small-scale models [62] or numerical simulations [63] (Appendix C.18), this study focuses on real-scale buildings and their spatiotemporal variations [64], revealing temperature differences in component contribution up to -3.5°C during the day and 1°C at night.

Contrary to the prevailing assumption that greater thermal mass always enhances energy efficiency, the findings reveal that the spatiotemporal performance of building components influences its effectiveness. As detailed in Table 2, northern floor slabs positively influence indoor temperatures during the day (Floor slab km factor: $1.62 \uparrow$) but negatively impact them at night (Floor slab km factor: $-0.38 \downarrow$), whereas southern internal walls have a positive effect both during the day (Internal wall km factor: $0.23 \uparrow$) and night (Internal wall km factor: $0.10 \uparrow$). To enhance thermal mass effectiveness, the study introduces cost-effective spatial modifications such as reducing northern internal walls to enlarge room volume, providing energy savings of 3.35–3.78 % in dormitories and 2.13–4.22 % in teaching buildings, as shown in Table 3.

4.3. Tailored thermal mass layout strategies for operational schedules

Building on these findings, this study proposes a novel strategy that shifts the focus from individual component innovation to spatial reconfiguration of existing thermal mass according to building schedules. Unlike prior work focusing on new materials and structures [65] (e.

g., PCMs [66], ESIWs [67]), this approach repurposes inherent mass (e. g., internal walls, slabs), achieving 7–8 % annual energy savings in cold-region campuses with high solar radiation. These findings highlight the need for schedule-specific spatial optimization of thermal mass.

Furthermore, aligning thermal mass use with operational schedules proves important. As shown in Fig. 10, dormitories benefit from daytime solar heat storage and nighttime release, yielding up to 4 % savings. In contrast, reducing heat storage improves efficiency in teaching buildings, with savings of up to 5 %. For nighttime-use dormitories, the tailored strategies yield 3–7 % energy savings, while misapplying these strategies to daytime-use teaching buildings increases energy use by 1–20 %, highlighting the role of operational schedules in thermal mass efficiency.

4.4. Expanding the application potential of the proposed strategies to broader fields

To extend the applicability of the proposed strategies, this study evaluates their performance across a range of broader scenarios. Appendix C and below provide detailed results. For example, this study may make up for the limitations of short-term decarbonization decisions that risk exceeding carbon budgets [68] by leveraging the energy-saving potential of inherent thermal mass, especially relevant for vulnerable communities that often lack access to emerging technologies or the ability to afford high upfront costs [69]. They can also be integrated with the design of equipment and building structures [70], and when combined with distributed building-integrated photovoltaic (BIPV) systems, they enable thermal mass-driven storage and utilization in different

building functions [71] and more effectively regulate renewable energy supply [72]. Moreover, these strategies extend the conventional focus on solar radiation from external façades and rooftops to optimize indoor spaces [73], making better use of underutilized interior surfaces such as walls and floors. Within this context, the proposed strategies offer notable potential to support sustainable urban development.

As shown in Fig. 12, evaluation results confirm consistent energy-saving performance and highlight optimization opportunities across varied building development stages and advanced strategies. The proposed thermal mass strategies exhibit favorable lifecycle economics when using conventional materials. For new buildings (Stage 1) in Fig. 12(a, b), they yield the highest returns. For partial renovations (Stage 2), reusing existing components enables cost recovery within 30 years for daytime-use buildings and 20 years for nighttime-use buildings. In full renovations (Stage 3), benefits decline: daytime-use buildings recover only ~65 % of the cost benefits observed in new constructions, with payback periods exceeding 70 years; nighttime-use buildings recover ~80 % within 50 years. For buildings nearing the end of life (≤ 30 years for daytime use, ≤ 20 years for nighttime use), major retrofits are not cost-effective. As demonstrated in Fig. 12(c), applying PCMs [59] to components with positive thermal response significantly improves energy savings. Specifically, PCMs are most effective in southern internal walls and floor slabs for dormitories, and in northern floor slabs and southern walls for teaching buildings. These results confirm the technical and economic viability of combining the proposed strategies with sustainable urban development.

4.5. Limitations and future work

This study still has limitations that warrant future research. First, this paper primarily focuses on heating consumption, a major contributor to urban building energy consumption, indicating the need for further exploration of thermal mass optimization strategies tailored to different climate zones (see Appendix C.19). Second, this study examines buildings with daytime and nighttime heating demands, while different

building types follow distinct and more specific usage patterns, which need further exploration (see Appendix C.19). Lastly, this study considers only the most commonly used building materials. As new materials and structural innovations emerge, integrating findings from the present study with ongoing research (see Fig. 12) could unlock broader opportunities for enhancing energy efficiency.

5. Conclusion

The substantial benefits of thermal mass in building components, particularly in dense urban environments, remain underutilized. As ongoing urbanization drives increased construction and energy demand, untapped thermal mass and stored energy continue to grow, highlighting their rising potential. A notable issue is the misalignment between thermal mass storage-release cycles and building schedules, leading to inefficient energy exchanges occurring daily. However, tailoring thermal mass strategies to specific building demands is nontrivial due to limited focus on this misalignment, a lack of methods to address the accompanying spatial and temporal variability in the thermal behavior of building components, and limited access to large-scale, detailed, real-world thermal mass behavior data. These challenges further complicate optimization efforts. In response, measured data from 76 rooms were analyzed to identify spatiotemporal variations and impact factors in component thermal behavior, guiding the development of optimization strategies that were subsequently validated through year-round simulations across cold regions.

This study aims to synchronize thermal mass storage-release cycles with building-specific energy demands, proposing spatial optimization strategies that enhance energy efficiency. The proposed strategies extend the benefits of thermal mass from individual components to larger building systems, complementing materials and structural design research advancements. As future urban sustainable development introduces more diverse energy-use scenarios and advanced technologies, the proposed strategies can be integrated with them and consistently deliver benefits by utilizing previously underused building internal

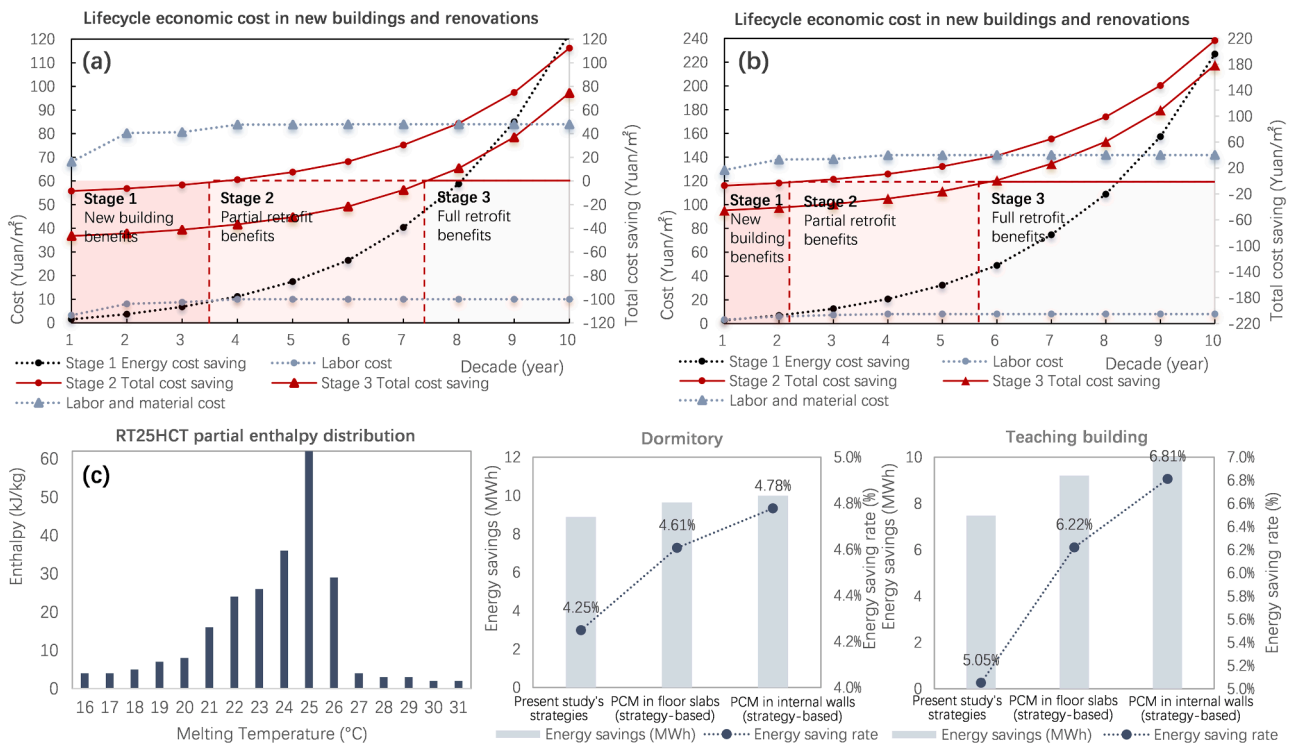


Fig. 12. Building material utilization and energy-saving effects: Lifecycle economic cost analysis of the proposed strategies using conventional materials in (a) daytime-use buildings, (b) nighttime-use buildings, and (c) integration with advanced materials such as PCMs.

surfaces. Specifically, our findings are threefold: (i) Identifying thermal mass optimization potential: Evaluating the alignment of component contributions with building operational schedules reveals optimization potential, as temperature variations of -3.5 to 1°C occur throughout the day; (ii) Proposing tailored strategies from mechanism analysis: Measurements inform optimization steps and simulations suggest energy saving potential, achieved through thermal mass redistribution and schedule adjustments. For daytime-use buildings, reducing heat storage in thermal mass is recommended, while for nighttime-use buildings, increasing daytime heat storage for nighttime use is suggested; and (iii) Validating broad applicability: Strategies in individual and grouped buildings achieve energy savings of 4–12 % across cold regions, with greater benefits observed in areas with higher solar radiation.

CRediT authorship contribution statement

Yichen Han: Writing – original draft, Investigation, Conceptualization. **Zhengyu He:** Writing – review & editing, Software, Formal analysis. **Shuangdui Wu:** Software, Methodology. **Yuqiu Liu:** Investigation, Data curation. **Yingkai Lian:** Investigation. **Chaohong Wang:** Investigation. **Jiajia Feng:** Investigation. **Zhengnan Zhou:** Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.adapen.2025.100224](https://doi.org/10.1016/j.adapen.2025.100224).

Data availability

Data will be made available on request.

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